

José Ferreira

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EDUCATION

PhD in Physics

University of Aveiro

Supervisors: Miguel Zilhão, Carlos Herdeiro and Raimon Luna

08/2023 - PRESENT

MSc in Physics

Faculty of Sciences of the University of Lisbon

Supervisors: Nelson Nunes and Tiago Barreiro

09/2020 - 09/2022

BSc in Physics

Faculty of Sciences of the University of Lisbon

09/2017 - 07/2020

RESEARCH INSTITUTES

Gr@v

Member of the Gravitation group at the *Center for Research & Development in Mathematics and Applications* of the University of Aveiro, Portugal.

08/2023 - PRESENT

Institute of Astrophysics and Space Sciences

Member of the cosmology group at the *Institute of Astrophysics and Space Sciences*, Lisbon node, Portugal.

09/2021 - 07/2023

TEACHING

Assistant

University of Aveiro

Courses: Numerical and Statistical Methods

09/2022 - PRESENT

Assistant

Faculty of Sciences of the University of Lisbon

Courses: Experimental Physics I, Elements of Physics, Mathematical Methods for Physicists, General Physics

02/2022 - 07/2023

PUBLICATIONS

Jordan Nicoules, **José Ferreira**, Carlos A. R. Herdeiro, Eugen Radu, and Miguel Zilhão. “Splitting the Gravitational Atom: Instabilities of Black Holes with Synchronized/Resonant Hair”. [arXiv: 2509.20450 \[gr-qc\]](https://arxiv.org/abs/2509.20450).

José Ferreira, Tiago Barreiro, José Mimoso, and Nelson J. Nunes. “Testing Λ -free $f(Q)$ cosmology”. *Phys. Rev. D* 108 (6 Sept. 2023), p. 063521. [10.1103/PhysRevD.108.063521](https://arxiv.org/abs/10.1103/PhysRevD.108.063521).

José Ferreira, Tiago Barreiro, José Mimoso, and Nelson J. Nunes. “Forecasting $F(Q)$ cosmology with Λ CDM background using standard sirens”. *Phys. Rev. D* 105 (12 June 2022), p. 123531. [10.1103/PhysRevD.105.123531](https://arxiv.org/abs/10.1103/PhysRevD.105.123531).

COMMUNICATIONS

TALKS

“Black Holes as Particle Accelerators”. 24th of October, 2025. Annual Meeting of PhD Students from Maths Department, University of Aveiro, Portugal (Links: [pptx](#))

“Splitting the (Gravitational) Atom”. 21th-25th of July, 2025. New Frontiers in NR 2025, Mallorca, Spain. (Links: [pptx](#), [pdf](#))

“(In)stability of Q-Hairy Black Holes”. 20th-22th of May, 2025. NewFunFICO Meeting, Valencia, Spain. (Links: [pptx](#))

“Non-linear Superradiance of Charged Black Holes: Beyond Spherical Symmetry”. 18th-22th of November, 2024. VI Amazonian Symposium on Physics, Belém, Brazil. (Links: [pptx](#), [pdf](#))

“Using Standard Sirens to test non-metric theories of gravity”. 19th-20th of December, 2023. XVI Black Holes Workshop, Porto, Portugal. (Links: [pptx](#), [pdf](#))

“Forecasting $F(Q)$ Cosmology with Λ CDM Background using Standard Sirens”. 4th-6th of May, 2022. Iberian Cosmology Meeting 2022, Barcelona, Spain. (Links: [recording](#), [pdf](#), [pptx](#))

SEMINARS

“Cosmological Tests on Non-Metricity Based Theories of Gravity”. 20th of September, 2023. Gr@v, Aveiro, Portugal. (Links: [recording](#), [pptx](#), [pdf](#))

POSTERS

“Splitting the (Gravitational) Atom”. 21th-25th of July, 2025. New Frontiers in NR 2025, Mallorca, Spain. (Links: [pdf](#))

“Testing Λ -Free $f(Q)$ Cosmology”. 3th-5th of May, 2023. Dark Matter and Stars, Lisbon, Portugal. (Links: [pdf](#))

> DIGITAL SKILLS

High Level Languages Julia, Python, Mathematica

Multimedia Editing GIMP, Photoshop, Inkscape, Kdenlive, Audacity

Low Level Languages C, Fortran, CUDA

Web Development Quarto, Hugo

I am an avid Linux user, with experience using it on personal machines, servers, clusters and microcontrollers. I work mostly on a terminal based ecosystem, creating and making use of terminal utilities to develop my projects.

👥 ACADEMIC ACTIVITIES

Supervisor at the IAstro Summer Internships 2023

07/2023

Institute of Astrophysics and Space Sciences

Supervisor of two undergraduate projects that consisted on constraining modified gravity cosmological models with current and forecast datasets coming from different sources.

Podcast “Physically Speaking”

12/2020 - 12/2022

[Fisicamente Falando](#) [Fisicamente Falando](#) [fisicamentefalando.nfef](#) [fisicamente_falando_](#)

A student driven podcast created during the pandemic where we interviewed faculty members to speak about their lives, both inside and outside of academia. I am in charge of the audio and video editing and also take part in some of the interviews.

Member of the Student Body for Physicists and Physical Engineers

10/2020 - 10/2021

[nfef-fcul.com](#) [NFEFFCUL](#) [nfeffcul](#)

I was in charge for the development of the digital tools of the student body of physicists and physical engineers. This included the website, the online repository (which is made up of student provided content) and the Discord channel.

📡 OUTREACH

European Researchers’ Night 2025

26/09/2025

Coordinator of the participation of Gr@v in the European Researcher’s Night in Aveiro, Portugal. Using digital content, we explained the physics behind Gravitational Waves and lensing around compact objects.

“Echoes and Shadows: How to Detect Black Holes”

03/05/2025

[Presentation \(.pptx\)](#)

A Presentation, for high school students, on what are black holes and how to find them.

Project “Physics on the loose”

01/2023 - 07/2023

A project, hosted at the Faculty of Sciences of the University of Lisbon, with a room full of interactive experiments to illustrate several fundamental concepts in Physics.

“When a Black Hole Falls Inside... Another Black Hole”

15/04/2023

[Presentation \(.pptx\)](#)

Presentation of an Ignite style talk in an event organized by the Institute of Astrophysics and Space Sciences that brings space sciences to cities far away from the main urban areas.